

Developing a Research Topic

14.33 Research and Communication in Economics

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September 14, 2026

Agenda

1. Respond to syllabus feedback and reiterate AI disclosure policy
2. Recap of previous lecture: Research topic as question + setting + data
3. Recap of causal inference
4. Examples of topics motivated by question vs. setting vs. data
5. Advice on spotting questions, settings, and data

Recap: question + setting + data

- **Question:** what your paper will answer
 - Motivated by economic theory or a policy issue
- **Setting:** institutional details that enable a causal comparison or test of a theory
- **Data:** measurements used to answer the question in the setting
- You will iterate among these components as you learn more about each one
- Examples last time: housing development (causal) and private information in insurance (descriptive)

Recap of causal inference

Causal inference basics

- A causal **question** asks what *would* happen to outcome Y if treatment D changed
- In the **data**, we can observe only what *actually* happens for each unit i
- We address the missing counterfactual with assumptions based on institutional knowledge of the **setting**
 - Free, detailed, and approachable online textbook: [Causal Inference: The Mixtape](#)
 - Animated visualizations of some approaches: [Nick Huntington-Klein's website](#)

The fundamental problem of causal inference

- Notation: potential outcomes for unit i (at time t) are $Y_i(1)$ when $D_i = 1$, $Y_i(0)$ when $D_i = 0$

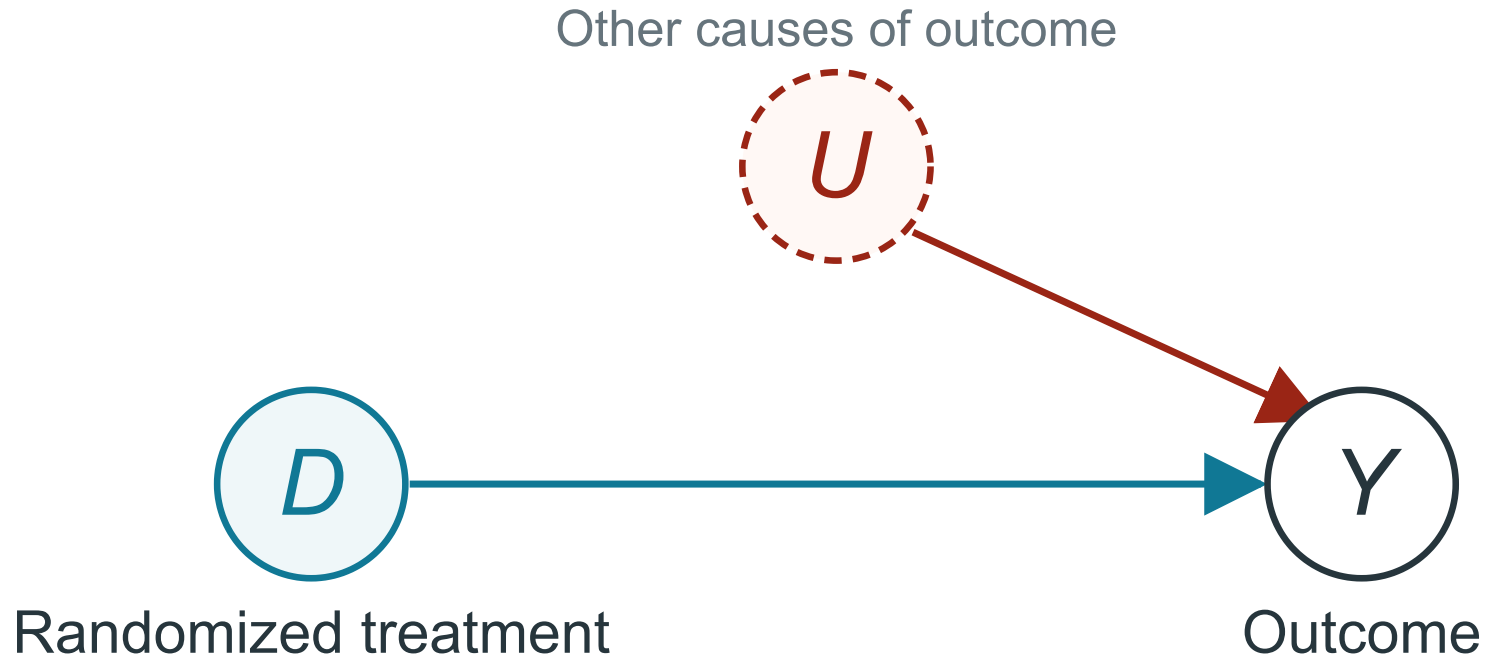
Realized treatment	$Y_i(1)$	$Y_i(0)$
$D_i = 1$	Observed	Missing
$D_i = 0$	Missing	Observed

- Problem: We cannot identify the individual causal effect $Y_i(1) - Y_i(0)$
- Solution: An identification strategy states the assumptions that let us learn a causal effect from observed data

Selected strategies for identifying causal effects

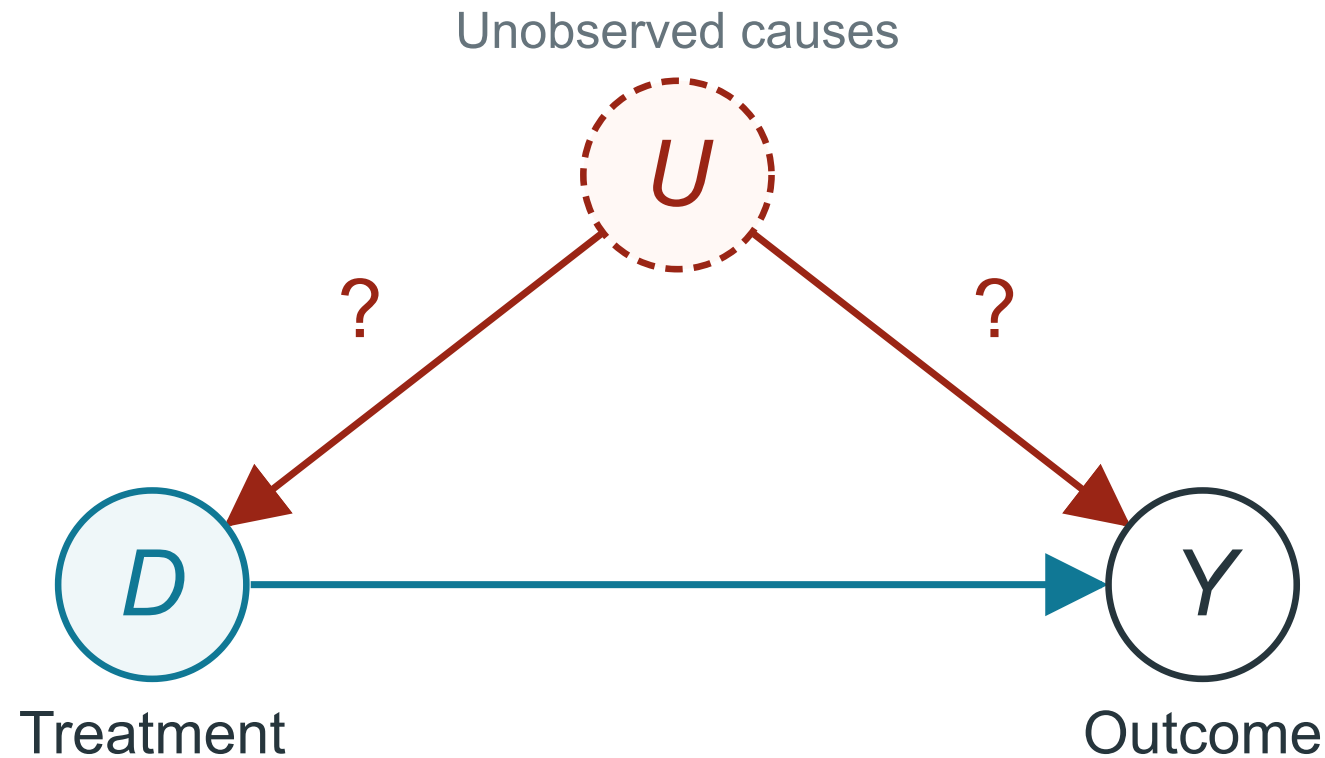
1. Randomized controlled trial
2. Selection on observables
3. Instrumental variables
4. Regression discontinuity
5. Bunching
6. Difference-in-differences

Randomized controlled trial



- Random assignment makes the groups comparable on average
- The untreated-group mean estimates the treated group's counterfactual mean

Observational data: what else determines treatment?



- D and Y may share unobserved causes, U
- Association combines the causal path with $D \leftarrow U \rightarrow Y$

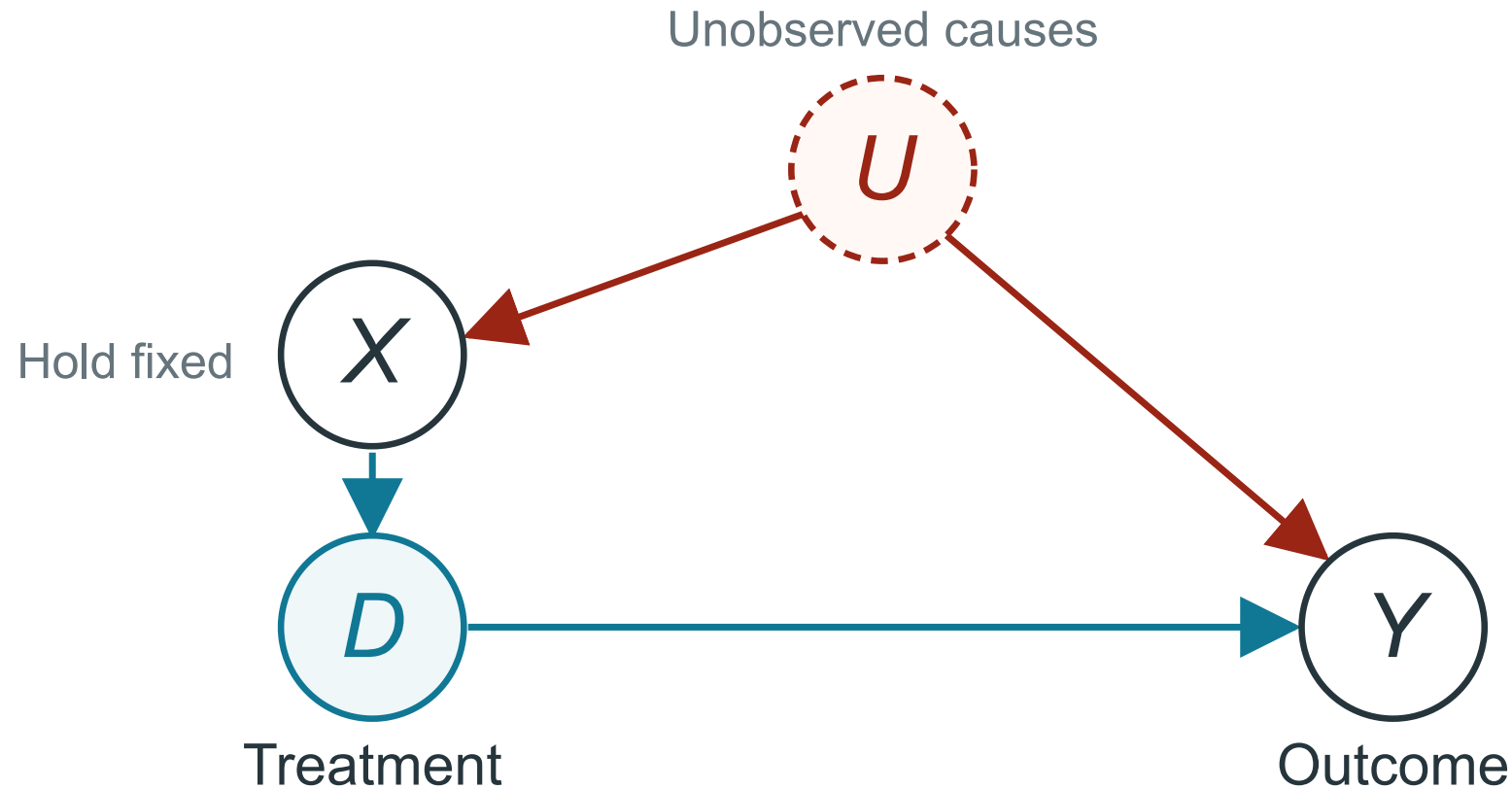
Exercise: interrogating causal relationships

The Economist, “Is AI putting graduates out of work already?”



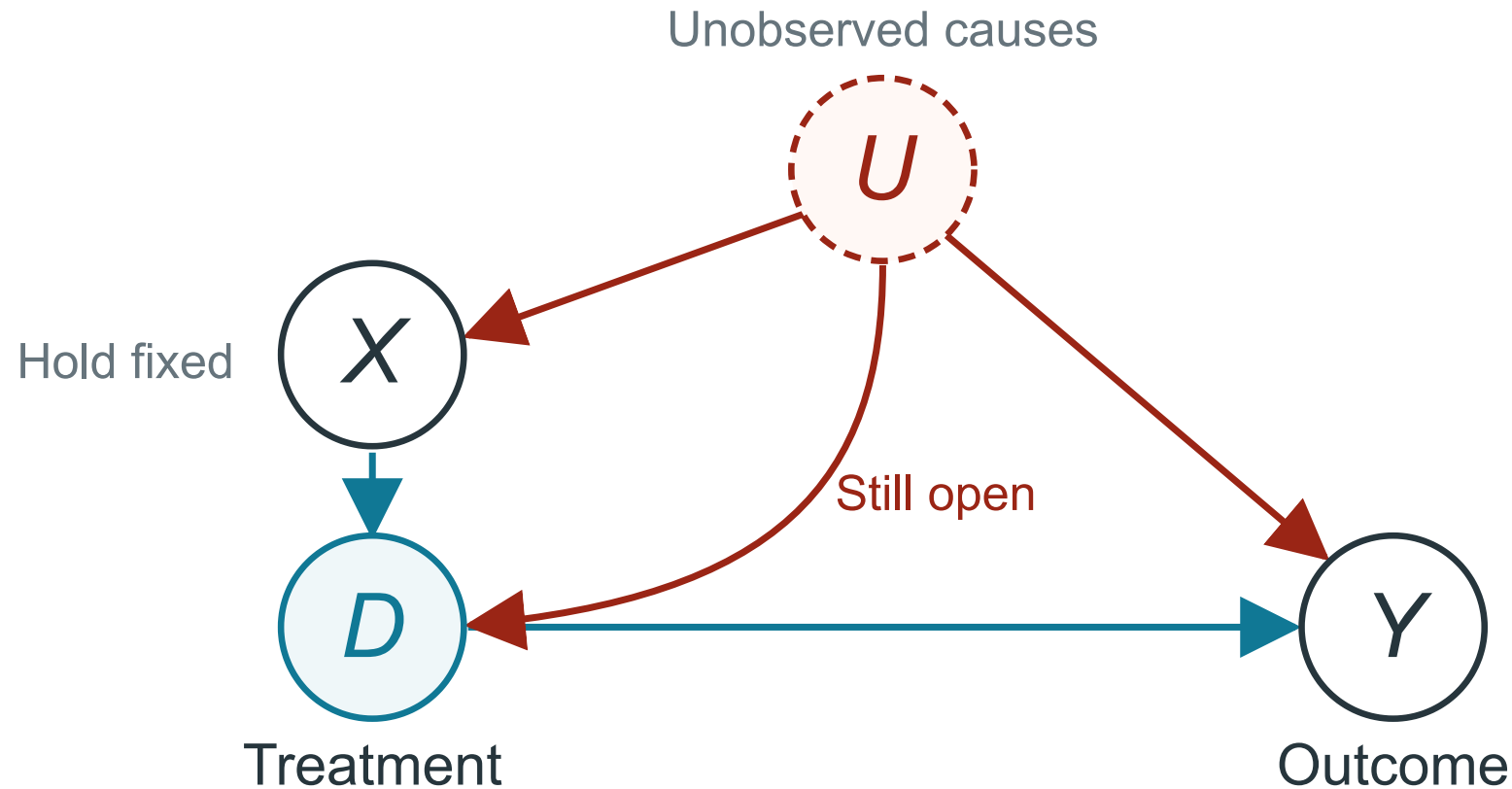
- State the causal relationship that the graph's title implies.
- State alternative causal explanations consistent with the data.

Observational design #1: selection on observables



- Compare treated and untreated units with the same pre-treatment X
- Assumption: comparing within X blocks the confounding “backdoor” path from D to Y

A control is not automatically enough

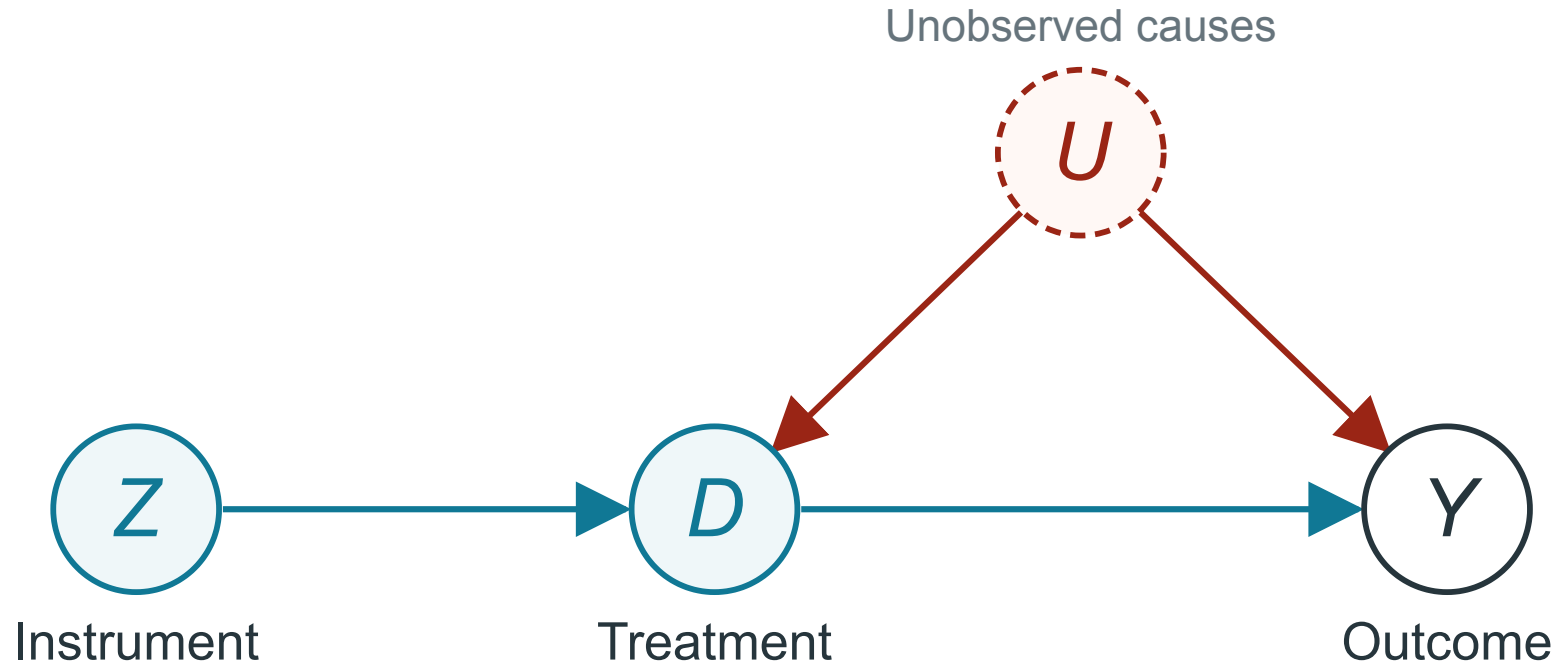


Key question: Why is variation in D as good as random within X ?

Advice: pursuing selection on observables

- Setting as motivation: treatment is assigned systematically but you observe the relevant inputs
 - **Abdulkadiroğlu, Angrist, Narita, and Pathak (2017)**: centralized school assignment uses priorities and preferences that the researcher observes
 - **Borusyak and Hull (2023)**: Chinese high-speed rail proposals had equally-suitable connections but only some were built
- Data as motivation: language models can represent high-dimensional, nonstandard data
 - **Bajari et al. (2025)**: encode product text and images as quality controls in a hedonic price index

Observational design #2: instrumental variables



- Focus on the variation in D induced by Z
- Assumptions:
 1. First stage: Z changes D
 2. Independence: Z and Y have no common causes
 3. Exclusion: Z affects Y only through D

Advice: pursuing instruments

- Setting as motivation: know one specific observable factor that changes treatment
 - **Geography, weather, or institutional quirks**
 - **Decision-makers with different tendencies**, with as-if-random assignment to the decision-maker

Cunningham, “Five Common Distinct IV Designs”

Observational design #3: regression discontinuity

$$D_i = 1[X_i \geq c]$$

$$\tau_{RDD} = \lim_{x \downarrow c} E[Y_i \mid X_i = x] - \lim_{x \uparrow c} E[Y_i \mid X_i = x]$$

- Treatment changes at cutoff c
- Assumption: average untreated potential outcomes are continuous at c

Advice: pursuing regression discontinuity

- Setting as motivation: find a stark cutoff rule that changes treatment
 - **Lee (2008)**: narrowly winning an election changes whether a party holds office
- Learn exactly how the running variable and cutoff determine treatment
- Ask whether units can sort around the cutoff or whether other rules change there

Exercise: a tax notch and a regression discontinuity

Los Angeles' mansion tax has a sharp notch: no tax below a \$5.4 million sale price and a 4% tax on the full price above it.

3 minutes: Is the threshold a useful cutoff for an RDD?

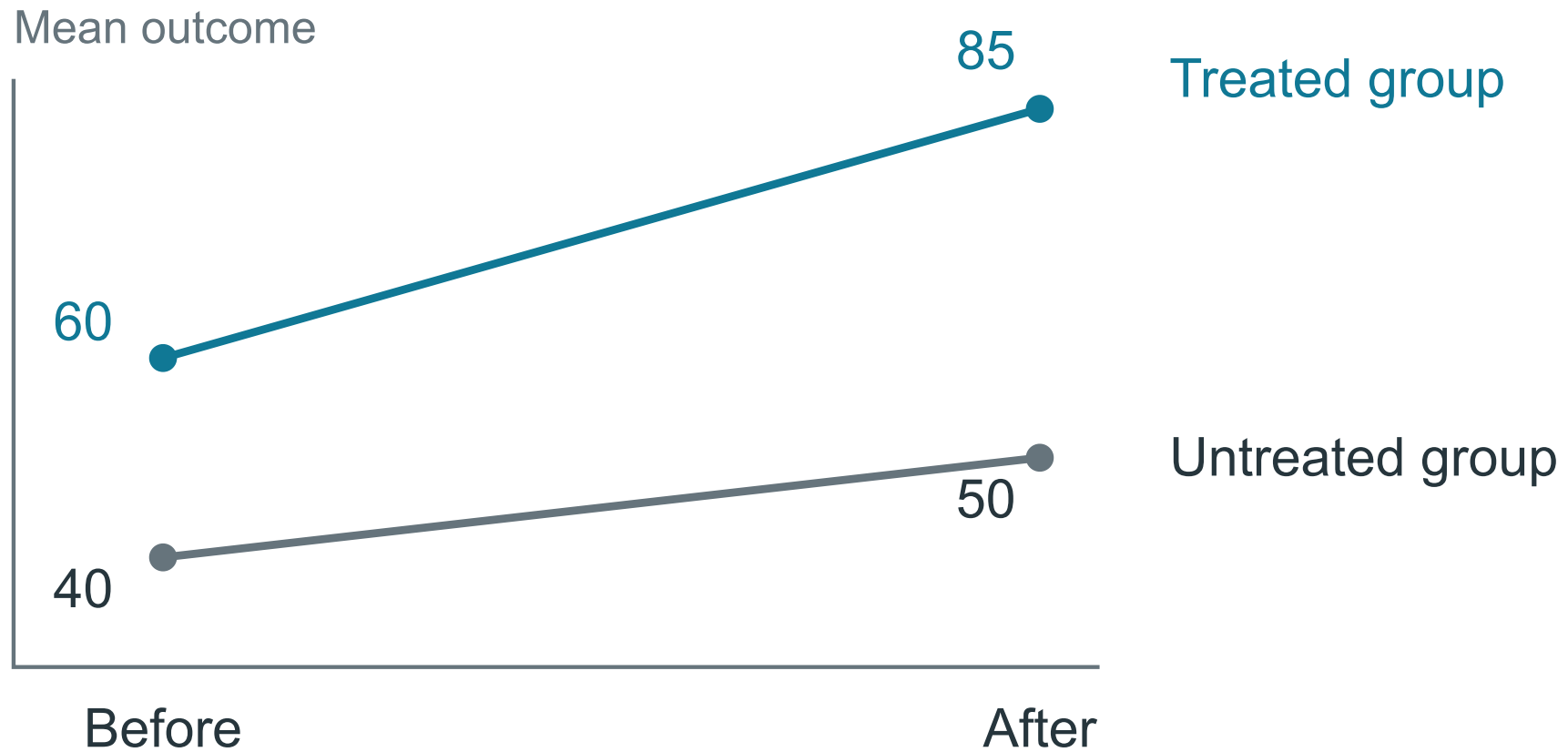
Observational design #4: bunching

- If people can sort around a threshold, an RDD comparison is not clean
- Make lemons into lemonade: the sorting itself is informative
- Infer the smooth **counterfactual density** without the notch
- **Bunching:** use excess mass near the cutoff to measure behavioral responses

Observational design #5: difference-in-differences

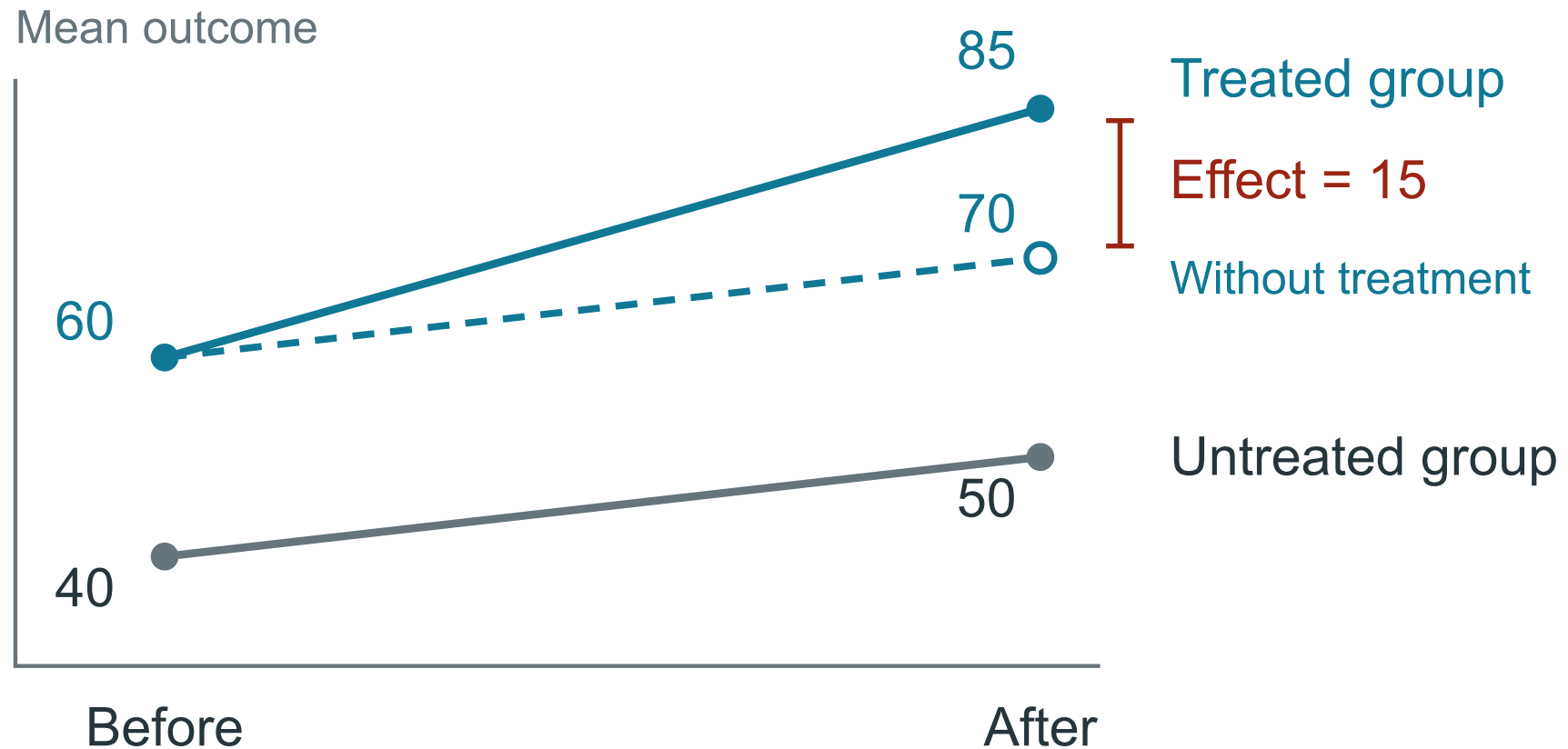
- Observe treated and untreated groups before and after treatment begins
- Group levels may differ before treatment
- Use the untreated group's change to infer the treated group's counterfactual change

Difference-in-differences: what we observe



Hypothetical numbers; treatment begins between periods.

Difference-in-differences: infer the missing trend



Hypothetical numbers; treatment begins between periods.

Parallel trends assumption: the untreated potential outcomes of both groups change similarly, even if their levels are different

Advice: pursuing difference-in-differences

- Setting as motivation: a policy changes for some groups but not others or rolls out gradually
- Similar pre-treatment trends are useful evidence, not proof
- Institutional knowledge must support the comparison group and timing

Forming a research topic

Economics is a broad field

- Be creative in seeing economic concepts
- **Examples of economic research across a variety of fields**

Three ways to start

- Research is a nonlinear and iterative process
- The following slides highlight papers that plausibly came from different starting points:
 - **Question-led:** something you want to understand
 - **Setting-led:** an institutional detail that makes evidence informative
 - **Data-led:** an opportunity to measure something previously hard to see

Question-led: does Nash equilibrium predict behavior?

- **Starting question:** Do the predictions of Nash equilibrium apply in the real world?
- **Your task:** Find behavior that reveals players' strategies and payoffs
- **Hint:** What is a relevant setting with *players* and *games*?

Question-led: mixed strategies at Wimbledon

- **Question:** test Nash equilibrium in a real-world game
- **Setting:** tennis servers and receivers choose left or right simultaneously
 - A mixed-strategy Nash equilibrium predicts equal success rates across actions used with positive probability
- **Data:** Wimbledon match videos
- **Walker and Wooders (2001), “Minimax Play at Wimbledon”**
- Broader lesson: Economics is not only what appears in the newspaper’s business section

Setting-led: where can a bridge be built?

- A tributary increases downstream river flow and bridge-construction costs
- Bridges are more common just upstream of confluences
- **Your task:** What economic question could this variation answer?
- **Hint:** Why do we care about variation in bridge access?

Setting-led: bridges and economic development

- **Setting:** river geography creates variation in bridge access
- **Question:** what is the economic impact of transportation infrastructure investment?
- **Data:** river geography and bridge locations
- **Tompsett (2026), “Bridges”**
- Broader lesson: Expertise outside economics is useful

Data-led: local housing permits

- **D'Amico, Soltas, and Wang (2026)** collect housing permit records across 60 U.S. cities, 2000–2025
- Measures the time from a permit application to completion
- **Your task:** Why could we care about these data?

Data-led: where is housing slow to build?

- **Data:** newly harmonized data on permitting timelines
- **Question:** is housing slower to build in high-cost places?
- **Setting:** developers must apply for permits, and approval processes differ across cities
- Broader lesson: Harmonizing scattered public records can benefit both your project and other researchers

Finding inspiration

Start with what you already know

- Internships, hometowns, hobbies, and historical knowledge
- Notice rules: prices, queues, eligibility, assignments, and contracts

Review unique data aggregators

- Proprietary datasets: [Dewey Data](#)
- Unusual datasets: [Data Is Plural](#)
- Freedom of Information Act releases: [Data Liberation Project](#)

Read the newspaper with a discerning eye

- **Question:** why does this story matter?
- **Setting:** is there a rule, reform, or incentive?
- **Data:** does the story identify an unusual measurement?
- Major outlets: [WSJ](#), [NYT](#), [FT](#), [Bloomberg](#)
- Long-form discovery: [Longreads](#); investigative reporting: [ProPublica](#)
- Specialized outlets: [Governing](#), [State Newsroom](#), [Utility Dive](#), [Waste Dive](#)

Boston Globe case study: headline

Boston Globe: “This program is driving up gas bills in Massachusetts. Did it make customers safer?”

This program is driving up gas bills in Massachusetts. Did it make customers safer?

By **Kate Selig** Globe Staff, Updated September 13, 2026, 1:25 p.m.



Boston Globe case study: question

Massachusetts gas customers have [spent millions each year on a state program](#) incentivizing utilities to replace old, leaky pipes in the name of safety and [reducing greenhouse gas emissions](#).

But with energy bills skyrocketing, state legislators are considering [scaling back or outright eliminating](#) the program, which the Senate estimates could save consumers more than a billion dollars on their gas bills.

Now, a new study gives opponents of the program further ammunition. A [report](#) from The Future of Heat Initiative, a nonprofit funded by climate-focused foundations, has found that increased spending on pipe replacement has not measurably improved safety.

Indeed, the report concludes, states like Massachusetts that have aggressively pursued these projects have had more incidents than those that didn't, suggesting that digging up old pipes is itself a source of risk.

"You can feel confident in reducing the level of spending on pipe replacement programs without compromising the safety of the system," said Jamie Van Nostrand, the nonprofit's policy director and former chair of the Massachusetts Department of Public Utilities.

FEATURED VIDEO

The gas industry disputed the findings, arguing that pipeline replacement efforts have protected customers, and the failures those efforts have prevented from occurring cannot be measured.

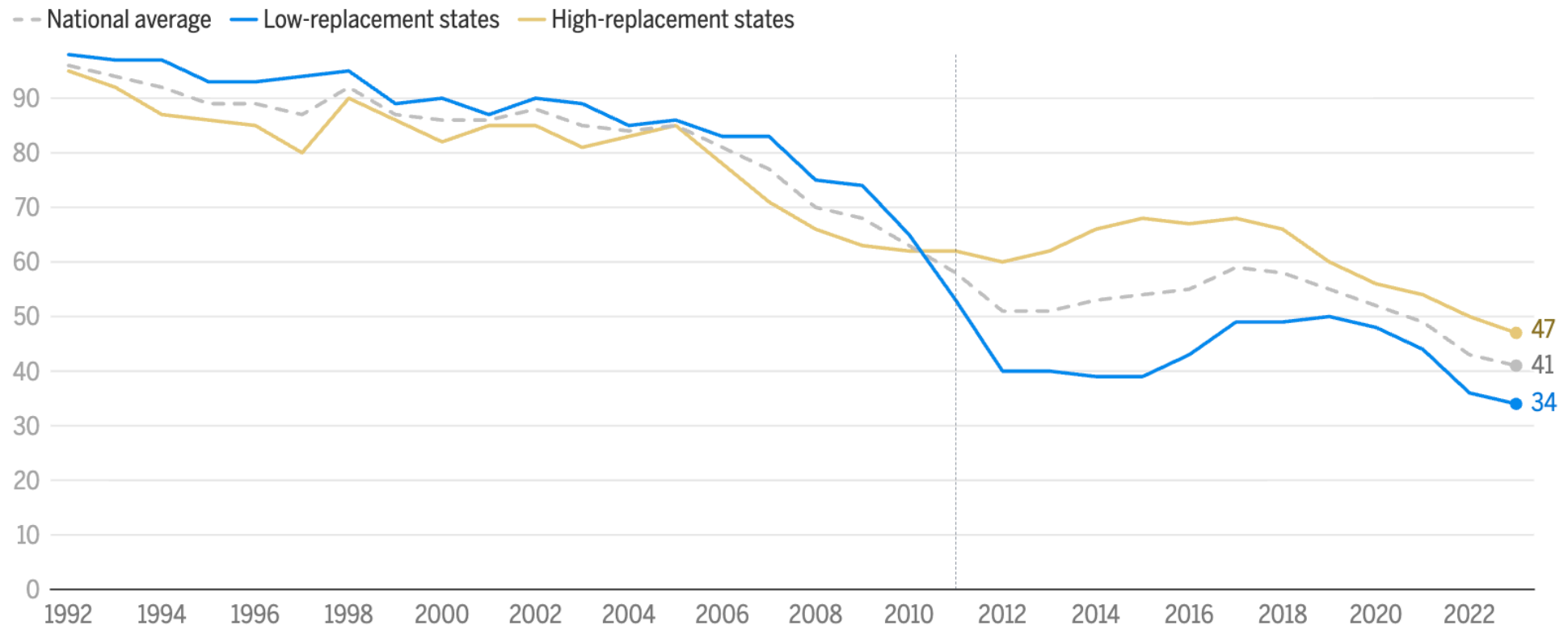
Related

■ **Step by step, Beacon Hill inching closer to fixing skyrocketing utility bills**

Boston Globe case study: data

States that replaced more pipe experienced a higher rate of significant incidents than low-replacement states after 2011

The chart below shows the five-year rolling average of reported incidents per million miles of gas distribution main.



Note: High-replacement states refers to the top 12 states replacing the greatest quantity of cast iron and bare steel mains from 1990 to 2025 (NY, PA, NJ, TX, OH, MA, CA, MI, FL, AL, IL, IN). Low-replacement states refers to all other states.

Chart: TAYLOR JOHNSTON/GLOBE STAFF • Source: [FoHI analysis of incident data from PHMSA](#)

Boston Globe case study: complementary reporting

WBUR: “Mass. orders utilities to spend less ratepayer money on natural gas pipelines”

Several other states with similar mechanisms for gas pipeline repairs, including [Illinois](#) and [Maryland](#), have also sought to rein in their programs in recent years.

Massachusetts state Sen. Mike Barrett, who co-chairs the legislative committee that oversees utilities, said one of the biggest problems is that companies have opted to replace leaking pipelines when less-costly repairs would have sufficed. Gas utilities make a profit when they build or repair infrastructure, so the more money they spend on pipelines, the more money they make.

Exercise: refine your preliminary idea list

- For one idea, propose a causal identification strategy or a test of economic theory
- Present the idea's "bull case" and "bear case" to a classmate
- Ask your classmate to suggest another strategy or data source

For Wednesday, September 16

- Complete the one-on-one meeting agenda in GitHub by 10:30 a.m. tomorrow